Masked observations

1 Masked observations

Theorem 1.1. *For every natural number t,* $\mathrm{whitecount}(t, 0) = 0$.

Proof. The goal follows from $\mathrm{whitecountzero}(t)$. □

Theorem 1.2. *For every natural number t and natural number n, if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n)$.

Proof. The goal follows from $\mathrm{whitecountblackstep}(t, n)$. □

Theorem 1.3. *For every natural number t and natural number n, if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n) + 1$.

Proof. The goal follows from $\mathrm{whitecountwhitestep}(t, n)$. □

Theorem 1.4. *For every natural number $t1$ and natural number $t2$ and natural number n, if* $\mathrm{centerwindowagree}(t1, t2, n)$*, then* $\mathrm{leftprefixagree}(t1, t2, n)$ *if and only if* $\mathrm{whitetraceagree}(t1, t2, n)$.

Proof. The goal follows from $\mathrm{prefixagreementiffwhitetrace}(t1, t2, n)$. □

Theorem 1.5. *For every natural number T and natural number P and natural number t and natural number n, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then there exists a natural number a such that there exists a natural number b such that $a < b$ and $b \leq$* $\mathrm{twopow}(\mathrm{whitecount}(t, n)) \wedge \mathrm{leftprefixagree}(t + a \cdot P, t + b \cdot P, n)$.

Proof. The goal follows from $\mathrm{compressedprefixrepeat}(T, P, t, n)$. □

Theorem 1.6. *For every natural number T and natural number P and natural number t and natural number n and natural number M, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then if* $\mathrm{prefixseparated}(t, P, n, M)$*, then* $M \leq \mathrm{twopow}(\mathrm{whitecount}(t, n))$.

Proof. The goal follows from $\mathrm{separatedprefixbound}(T, P, t, n, M)$. □

Theorem 1.7. *For every natural number T and natural number P and natural number t and natural number n, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $n < t + \mathrm{twopow}(\mathrm{whitecount}(t, n)) \cdot P$.

Proof. The goal follows from $\mathrm{edgeinformationbound}(T, P, t, n)$. □

Theorem 1.8. *For every natural number n and natural number t and natural number w and natural number P, if $n < (w+1) \cdot P$, then $n < t + \mathrm{twopow}(w) \cdot P$.*

Proof. The goal follows from $\mathrm{informationboundaudit}(n, t, w, P)$. □

2 Central collision recurrence

Theorem 2.1. *For every natural number Q and natural number t and natural number r, if $\mathrm{leftadjacentdefect}(Q, t) = \mathrm{true}$, then if $\mathrm{center}(t + r) = \mathrm{true}$, then if $\mathrm{center}(t + r + Q) = \mathrm{true}$, then $\mathrm{nearcollision}(Q, t, r + 1, 2 \cdot (r + 1))$.*

Proof. The goal follows from $\mathrm{unitdefectforcesnearcollision}(Q, t, r)$. □

Theorem 2.2. *For every natural number T and natural number P and natural number m and natural number t, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq t$, then if $\mathrm{leftadjacentdefect}(m \cdot P, t) = \mathrm{true}$, then $\mathrm{nearcollision}(m \cdot P, t, P, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodicunitdefectforcesnearcollision}(T, P, m, t)$. □

Theorem 2.3. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + P \wedge \mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodiccentralcollisionwindow}(T, P, m, N)$. □

Theorem 2.4. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then there exists a natural number t such that $N \leq t$ and $\mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodiccentralcollisions}(T, P, m, N)$. □

Theorem 2.5. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then $\mathrm{recurrentcentralcollision}(m \cdot P, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodicrecurrentcentralcolumn}(T, P, m)$. □

Theorem 2.6. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number s such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*leftcollision$(Q, b+s, r+1)$.

Proof. The goal follows from blackwindowforcesleftcollision(Q, b, r). □

Theorem 2.7. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicleftcollisionwindow(T, P, m, N). □

Theorem 2.8. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then there exists a natural number t such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicleftcollisions(T, P, m, N). □

Theorem 2.9. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentleftcollision$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentleftcolumn(T, P, m). □

Theorem 2.10. *For every natural number Q and natural number t and natural number k,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true.

Proof. The goal follows from collisioncodenonzero(Q, t, k). □

Theorem 2.11. *For every natural number Q and natural number t and natural number k,* collisioncode$(Q, t, k) \leq 6$.

Proof. The goal follows from collisioncodebound(Q, t, k). □

Theorem 2.12. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* collisioncode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 2$ *and* (collisioncode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$ leftcolour$(Q, t, k - 1) = 2)$.

Proof. The goal follows from collisioncodecolours(Q, t, k). □

Theorem 2.13. *For every natural number Q and natural number t and natural number k,* advancingatdepth(Q, t, k) *if and only if* collisioncode$(Q, t, k) = 3$ *or* collisioncode$(Q, t, k) = 5$.

Proof. The goal follows from advancingcollisioncodes(Q, t, k). □

Theorem 2.14. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftdefect$(Q, t + 1, k) =$ leftdefect$(Q, t, k + 1)$.

Proof. The goal follows from collisiontransmitsleft(Q, t, k). □

Theorem 2.15. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftgatedifference$(Q, t, k) =$ false.

Proof. The goal follows from collisiongatesagree(Q, t, k). □

Theorem 2.16. *For every natural number Q and natural number t and natural number k, if $2 \leq k$, then* collisioncode$(Q, t, k) = 1 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t, k-2)$ *and* (collisioncode$(Q, t, k) = 2 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t+Q, k-2)) \wedge$(collisioncode$(Q, t, k) = 3 \to$ leftdefect$(Q, t+1, k-1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 4 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t + Q, k - 2))) \wedge$ (collisioncode$(Q, t, k) = 5 \to$ leftdefect$(Q, t + 1, k - 1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 6 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t, k - 2))$).

Proof. The goal follows from collisionrighttransporttable(Q, t, k). □

Theorem 2.17. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* collisioncode$(Q, t, k) = 3 \to$ leftcolour$(Q, t + 1, k-1) = 2$ *and if* collisioncode$(Q, t, k) = 5$*, then* leftcolour$(Q, t+1, k-1) =$ 1.

Proof. The goal follows from advancingrightcolourtable(Q, t, k). □

Theorem 2.18. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t + 1, k) =$ not(leftdefect$(Q, t, k + 1)$) *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from gatedifferenceiffflip(Q, t, k). □

Theorem 2.19. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k+1) =$ true*, then* leftdefect$(Q, t+1, k) =$ false *if and only if* leftgatedifference$(Q, t, k) =$ true*.*

Proof. The goal follows from incomingdefecterasediff(Q, t, k). □

Theorem 2.20. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k+1) =$ true*, then if* leftdefect$(Q, t+1, k) =$ false*, then* collisioncode$(Q, t, k) = 0$*.*

Proof. The goal follows from erasurehasnocollision(Q, t, k). □

Theorem 2.21. *For every natural number Q and natural number k, if $0 < k$, then* leftcolour$(Q, 0, k) = 0$ *or* leftcolour$(Q, 0, k) = 2$*.*

Proof. The goal follows from seedleftcolours(Q, k). □

Theorem 2.22. *For every natural number Q and natural number k, if $2 \leq k$, then* collisioncode$(Q, 0, k) = 0$*.*

Proof. The goal follows from seedleftcollisionzero(Q, k). □

Theorem 2.23. collisioncode$(1, 1, 2) = 5$*.*

Proof. The goal follows from seedadvancingcollisionbirth. □

Theorem 2.24. leftdefect$(3, 0, 3) =$ true *and* leftdefect$(3, 1, 2) =$ false $\wedge$ leftgatedifference$(3, 0, 2) =$ true*.*

Proof. The goal follows from seedgateerasure. □

Theorem 2.25. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k) =$ true*, then if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then* advancingleftcone(Q, t, k)*.*

Proof. The goal follows from leftdefectblackcone(Q, t, k). □

Theorem 2.26. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number s such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*advancingleftcollision$(Q, b+s, r+1)$*.*

Proof. The goal follows from blackwindowforcesadvancingcollision(Q, b, r). □

Theorem 2.27. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* advancingleftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicadvancingleftcollisionwindow(T, P, m, N). □

Theorem 2.28. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentadvancingleftcollision$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentadvancingleftcolumn(T, P, m). □

Theorem 2.29. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t, k) =$ false *if and only if* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 3$.

Proof. The goal follows from leftdefectfalsecolours(Q, t, k). □

Theorem 2.30. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t, k) =$ true *if and only if* leftcolour$(Q, t, k) = 1$ *or* leftcolour$(Q, t, k) = 2$.

Proof. The goal follows from leftdefecttruecolours(Q, t, k). □

Theorem 2.31. *For every natural number Q and natural number t and natural number k, if $t < k$, then* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 2$.

Proof. The goal follows from leftcoloursoutsideseedcone(Q, t, k). □

Theorem 2.32. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then $k \leq t + 1$.*

Proof. The goal follows from collisioncodesupport(Q, t, k). □

Theorem 2.33. *For every natural number Q and natural number t and natural number k,* gatecode$(Q, t, k) \neq 0$ *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from gatecodenonzero(Q, t, k). □

Theorem 2.34. *For every natural number Q and natural number t and natural number k,* $\mathrm{gatecode}(Q, t, k) \leq 6$.

Proof. The goal follows from $\mathrm{gatecodebound}(Q, t, k)$. □

Theorem 2.35. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* $\mathrm{gatecode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2)$.

Proof. The goal follows from $\mathrm{gatecodecolours}(Q, t, k)$. □

Theorem 2.36. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftdefect}(Q, t, k+1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, k)$.

Proof. The goal follows from $\mathrm{erasureeventofstep}(Q, t, k)$. □

Theorem 2.37. *For every natural number Q and natural number t and natural number k, if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 3$ *if and only if* $\mathrm{gatecode}(Q, t, k) \leq 3$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 2$ *or* $4 \leq \mathrm{gatecode}(Q, t, k)$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 1$.

Proof. The goal follows from $\mathrm{erasureblackphaseiff}(Q, t, k)$. □

Theorem 2.38. *For every natural number Q and natural number t and natural number k, if $t+1 < k$, then if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 0$.

Proof. The goal follows from $\mathrm{erasureoutsideseedconewhite}(Q, t, k)$. □

Theorem 2.39. *For every natural number Q and natural number t and natural number k, if $2 \leq k$, then if* $\mathrm{advancingatdepth}(Q, t, k)$*, then if* $\mathrm{leftdefect}(Q, t+k, 0) = \mathrm{false}$*, then there exists a natural number j such that* $\mathrm{firstpatherasure}(Q, t, k, j)$.

Proof. The goal follows from $\mathrm{advancingsourcefirsterasure}(Q, t, k)$. □

Theorem 2.40. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then there exists a natural number u such that there exists a natural number d such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* $\mathrm{erasureevent}(m \cdot P, u, d)$.

Proof. The goal follows from periodicerasurewindow(T, P, m, N). □

Theorem 2.41. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P), *then* recurrentsourceerasure$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentsourceerasure(T, P, m). □

Theorem 2.42. noperiod *if and only if* norecurrentcentralcollision.

Proof. The goal follows from noperiodiffnorecurrentcentralcollision. □

Theorem 2.43. norowcenterperiod *if and only if* norecurrentcentralcollision.

Proof. The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

Theorem 2.44. problemone *if and only if* norecurrentcentralcollision.

Proof. The goal follows from problemoneiffnorecurrentcentralcollision. □

Theorem 2.45. *If* norecurrentcentralcollision, *then* noperiod.

Proof. Assume h. The goal follows from noperiodofnorecurrentcentralcollision(h). □

Theorem 2.46. *For every natural number T and natural number P, if $0 < P$, then if* norecurrentcentralcollision, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofnorecurrentcentralcollision(T, P). □

Theorem 2.47. *If* norecurrentcentralcollision, *then* problemone.

Proof. Assume h. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision(h). □

Theorem 2.48. *For every natural number b and natural number d and natural number i,* pairphase(wittphi(b), wittxor(wittphi(d), wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$).

Proof. The goal follows from pairphasestep(b, d, i). □

Theorem 2.49. *For every natural number Q and natural number t and natural number k,* alignedphase$(Q, t, t + k + Q) =$ leftphase(Q, t, k).

Proof. The goal follows from alignedphaseleft(Q, t, k). □

Theorem 2.50. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t, k) = 0$ *if and only if* leftdefect$(Q, t, k) =$ false*.*

Proof. The goal follows from phasezeroiff(Q, t, k). □

Theorem 2.51. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t, k) = -1$ *or* leftphase$(Q, t, k) = 1$ *if and only if* leftdefect$(Q, t, k) =$ true*.*

Proof. The goal follows from phaseunitiff(Q, t, k). □

Theorem 2.52. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t+1, k) =$ leftphase$(Q, t, k+1)+$leftphaseflux(Q, t, k).

Proof. The goal follows from leftphasebalance(Q, t, k). □

Theorem 2.53. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftphaseflux$(Q, t, k) = -2 \cdot$ leftphase$(Q, t, k + 1)$*.*

Proof. The goal follows from collisionphaseflip(Q, t, k). □

Theorem 2.54. *For every natural number Q and natural number t and natural number k, if $0 < k$, then if* advancingatdepth(Q, t, k)*, then* leftphase$(Q, t+1, k-1) = -$ leftphase(Q, t, k)*.*

Proof. The goal follows from advancingphaseflip(Q, t, k). □

Theorem 2.55. *For every natural number Q and natural number t and natural number k,* gatecode$(Q, t, k) = 1 \to$ leftgatephase$(Q, t, k) = -1$ *and* (gatecode$(Q, t, k) = 2 \to$ leftgatephase$(Q, t, k) = -1) \wedge$ (gatecode$(Q, t, k) = 3 \to$ leftgatephase$(Q, t, k) = -1) \wedge$(gatecode$(Q, t, k) = 4 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 5 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 6 \to$ leftgatephase$(Q, t, k) = 1)$.

Proof. The goal follows from gatephasecases(Q, t, k). □

Theorem 2.56. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then* leftphase$(Q, t + 1, k) = 0$ *and* leftphaseflux$(Q, t, k) = -$ leftphase$(Q, t, k+1) \wedge$ (leftphase$(Q, t, k+1) = -1 \vee$ leftphase$(Q, t, k + 1) = 1)$*.*

Proof. The goal follows from erasurephasebalance(Q, t, k). □

Theorem 2.57. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then* leftcolour$(Q, t+1, k) = 3 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$leftphase$(Q, t, k+1) = -1$ *and* leftcolour$(Q, t+1, k) = 0 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$ leftphase$(Q, t, k+1) = 1$.

Proof. The goal follows from erasureresetproduct(Q, t, k). □

Theorem 2.58. *For every natural number Q and natural number t and natural number k and natural number n,* pathphase$(Q, t, k, n) =$ leftphase$(Q, t, k)+$ pathflux(Q, t, k, n).

Proof. The goal follows from pathphasebalance(Q, t, k, n). □

Theorem 2.59. *For every natural number Q and natural number t, if $0 < Q$, then* leftphase$(Q, t, t + Q) = 1$.

Proof. The goal follows from seedleftphase(Q, t). □

Theorem 2.60. *For every natural number Q and natural number t, if $0 < Q$, then* leftphase$(Q, t + (t + Q), 0) = 1 +$ pathflux$(Q, t, t + Q, t + Q)$.

Proof. The goal follows from seedcenterphasebalance(Q, t). □

Theorem 2.61. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then* pathflux$(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

Proof. The goal follows from periodicseedflux(T, P, m, N). □

Theorem 2.62. leftphase$(3, 0, 3) = 1$ *and* leftphase$(3, 1, 2) = 0 \wedge$leftphaseflux$(3, 0, 2) = -1 \wedge$ leftgatephase$(3, 0, 2) = 1$.

Proof. The goal follows from seedphasecancellation. □

Theorem 2.63. *For every natural number Q and natural number t and natural number k and natural number s, if* rightmostleft(Q, t, k)*, then if $s < k$, then if* centercleanthrough(Q, t, s)*, then* rightmostleft$(Q, t + s, k - s)$.

Proof. The goal follows from rightmostleftprofile(Q, t, k, s). □

Theorem 2.64. *For every natural number Q and natural number t and natural number k, if* leadingadvance(Q, t, k)*, then if* centercleanthrough(Q, t, k)*, then* firstpatherasure$(Q, t, k, k - 2)$ *and* erasureevent$(Q, t + (k - 1), 0) \wedge (\text{gatecode}(Q, t + (k-1), 0) = 2 \vee \text{gatecode}(Q, t + (k-1), 0) = 5) \wedge \text{center}(t + (k - 1)) = \text{false}$.

Proof. The goal follows from leadingfirsterasure(Q, t, k). □

Theorem 2.65. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b + r) =$ true*, then there exists a natural number s such that there exists a natural number k such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge$* leadingadvance$(Q, b + s, k)$.

Proof. The goal follows from blackwindowforcesleadingsource(Q, b, r). □

Theorem 2.66. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that there exists a natural number k such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k)$.

Proof. The goal follows from periodicleadingsourcewindow(T, P, m, N). □

Theorem 2.67. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that there exists a natural number k such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k) \wedge$ firstpatherasure$(m \cdot P, t, k, k - 2) \wedge (\text{gatecode}(m \cdot P, t + (k-1), 0) = 2 \vee \text{gatecode}(m \cdot P, t + (k-1), 0) = 5)$.

Proof. The goal follows from periodicleadingerasurewindow(T, P, m, N). □

Theorem 2.68. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentleadingerasure$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentleadingerasure(T, P, m). □

Theorem 2.69. *For every natural number Q and natural number R and natural number t and natural number k,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k)+$ leftphase$(R, t + Q, k)$.

Proof. The goal follows from leftphaselagadd(Q, R, t, k). □

Theorem 2.70. *For every natural number Q and natural number R and natural number t and natural number k and natural number n,* $\mathrm{pathflux}(Q + R, t, k, n) = \mathrm{pathflux}(Q, t, k, n) + \mathrm{pathflux}(R, t + Q, k, n)$.

Proof. The goal follows from $\mathrm{pathfluxlagadd}(Q, R, t, k, n)$. □

Theorem 2.71. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leadingadvance}(Q, t, k)$*, then if* $\mathrm{centercleanthrough}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+k, 0) = 3 \leftrightarrow \mathrm{leftgatephase}(Q, t+(k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = -1$ *and* $\mathrm{leftcolour}(Q, t+k, 0) = 0 \leftrightarrow \mathrm{leftgatephase}(Q, t + (k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = 1$.

Proof. The goal follows from $\mathrm{leadingphasereset}(Q, t, k)$. □

Theorem 2.72. *For every natural number Q and natural number t, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t + 1, 0) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, 0)$ *if and only if* $\mathrm{gatecode}(Q, t, 0) = 2$ *or* $\mathrm{gatecode}(Q, t, 0) = 5$.

Proof. The goal follows from $\mathrm{centererasureiffgate}(Q, t)$. □

Theorem 2.73. *For every natural number Q and natural number b and natural number n and natural number i and natural number j and natural number k, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{gatecode}(Q, b + i, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + i, 0) = 5$*, then if* $\mathrm{gatecode}(Q, b + k, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + k, 0) = 5$*, then* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

Proof. The goal follows from $\mathrm{whiteblockgateconvex}(Q, b, n, i, j, k)$. □

Theorem 2.74. *For every natural number Q and natural number b and natural number n and natural number i and natural number j and natural number k, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if* $\mathrm{centercleanthrough}(Q, b, n+1)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{erasureevent}(Q, b + i, 0)$*, then if* $\mathrm{erasureevent}(Q, b + k, 0)$*, then* $\mathrm{erasureevent}(Q, b + j, 0)$ *and* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

Proof. The goal follows from $\mathrm{whiteblockerasureconvex}(Q, b, n, i, j, k)$. □

Theorem 2.75. $\mathrm{leadingadvance}(4, 5, 2)$ *and* $\mathrm{centercleanthrough}(4, 5, 2) \wedge \mathrm{gatecode}(4, 6, 0) = 5$.

Proof. The goal follows from seedleadinggatefive. □

Theorem 2.76. $\mathrm{leadingadvance}(7, 7, 4)$ *and* $\mathrm{centercleanthrough}(7, 7, 4) \wedge \mathrm{gatecode}(7, 10, 0) = 2$.

Proof. The goal follows from seedleadinggatetwo. □

Theorem 2.77. leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

Proof. The goal follows from seedleadingblackreset. □

Theorem 2.78. *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase$(incomingBit, laterBit,$ transferphase$(incomingBi$ *incoming.*

Proof. The goal follows from transferrecover$(incomingBit, laterBit, incoming, gate)$. □

Theorem 2.79. *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase$(incomingBit, laterBit, incoming, gate) =$ 0 *if and only if incoming* $=$ recoverphase$(incomingBit, laterBit, 0, gate)$.

Proof. The goal follows from transferkernel$(incomingBit, laterBit, incoming, gate)$. □

Theorem 2.80. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t{+}1, k) =$ transferphase(leftcell$(t, k{+}1)$, latergate(Q, t, k), leftphase$(Q, t, k{+}1)$, leftgatephase(Q, t, k)).

Proof. The goal follows from phasetransfer(Q, t, k). □

Theorem 2.81. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate(Q, t, k), leftphase$(Q, t{+}1, k)$, leftgatephase(Q, t, k)).

Proof. The goal follows from phaserecover(Q, t, k). □

Theorem 2.82. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then not* leftgatephase$(Q, t, k) = 0$ *and* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k), 0,$ leftgatephase(Q, t, k)).

Proof. The goal follows from erasurerecovery(Q, t, k). □

Theorem 2.83. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number t such that $N \leq t$ and* erasureevent$(m \cdot P, t, 0) \wedge$ gatecode$(m \cdot P, t, 0) = g$.

Proof. The goal follows from periodicerasureorientation(T, P, m, N, g). □

Theorem 2.84. *For every natural number Q and natural number t and natural number k, if* leadingadvance(Q, t, k)*, then if* centercleanthrough(Q, t, k)*, then there exists a natural number s such that $s + 1 < k$ and* center$(t + s) =$ true*.*

Proof. The goal follows from leadingblackbeforearrival(Q, t, k). □

Theorem 2.85. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap(b, r)*, then if* leftadjacentdefect$(Q, b+ r) =$ true*, then there exists a natural number k such that $k \leq r + 1$ and* leadingadvance(Q, b, k)*.*

Proof. The goal follows from blackgapforcesinitialleading(Q, b, r). □

Theorem 2.86. *For every natural number Q and natural number t, if* leadingadvance$(Q, t, 2)$*, then if* centercleanthrough$(Q, t, 2)$*, then* center$(t) =$ true*.*

Proof. The goal follows from leadingdepthtwoblack(Q, t). □

Theorem 2.87. *For every natural number Q and natural number b and natural number r and natural number g, if $0 < r$, then if* centerwindowagree$(b, b+ Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap(b, r)*, then if* erasureevent$(Q, b + r, 0)$*, then if* gatecode$(Q, b + r, 0) = g$*, then there exists a natural number k such that $k \leq r + 1$ and* leadingadvance$(Q, b, k) \wedge$ firstpatherasure$(Q, b, k, k - 2) \wedge$ gatecode$(Q, b + (k - 1), 0) = g$*.*

Proof. The goal follows from blackgaperasureorigin(Q, b, r, g). □

Theorem 2.88. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number b such that there exists a natural number k such that $N \leq b$ and $k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k - 1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k - 1), 0) = g$*.*

Proof. The goal follows from periodicblackerasureorigin(T, P, m, N, g). □

Theorem 2.89. *For every natural number t,* rightcell$(t+1, 1) =$ bxor(center(t), bxor(rightcell$(t, 1)$, right

Proof. The goal follows from rightimportstep(t). □

Theorem 2.90. *For every natural number Q and natural number t, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{bxor}(\mathrm{rightdefect}(Q, t, 1), \mathrm{rightimportdifference}(Q, t))$.

Proof. The goal follows from $\mathrm{rightdefectimportbalance}(Q, t)$. □

Theorem 2.91. *For every natural number Q and natural number t, if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t+Q) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightphase}(Q, t+1, 1) = -\mathrm{rightphase}(Q, t, 1)$.

Proof. The goal follows from $\mathrm{blackresetsurvival}(Q, t)$. □

Theorem 2.92. *For every natural number Q and natural number t, if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t+Q) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t+1, 1) = -\mathrm{rightphase}(Q, t, 2)$.

Proof. The goal follows from $\mathrm{blackresetimport}(Q, t)$. □

Theorem 2.93. *For every natural number T and natural number P and natural number m and natural number N and truth value $before$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then there exists a natural number t such that $N \leq t$ and* $\mathrm{rightimport}(t) = before \wedge \mathrm{rightimport}(t + m \cdot P) = \mathrm{not}(before) \wedge \mathrm{rightdefect}(m \cdot P, t+1, 1) = \mathrm{not}(\mathrm{rightdefect}(m \cdot P, t, 1))$.

Proof. The goal follows from $\mathrm{periodicrightimportorientation}(T, P, m, N, before)$. □

Theorem 2.94. *For every natural number Q and natural number t, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightimportdifference}(Q, t) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t+1, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t))) \cdot \mathrm{rightphase}(Q, t, 2)$.

Proof. The goal follows from $\mathrm{rightheadbirth}(Q, t)$. □

Theorem 2.95. *For every natural number Q and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q, start, finish)$*, then if* $\mathrm{centercleanthrough}(Q, start, finish - start)$*, then* $\mathrm{rightphase}(Q, finish, 1) = \mathrm{centersign}(start, finish - start) \cdot \mathrm{rightphase}(Q, start, 2)$.

Proof. The goal follows from $\mathrm{headimportpathphase}(Q, start, finish)$. □

Theorem 2.96. *For every natural number Q and natural number t, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t+1, 0) = \mathrm{false}$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{leftphase}(Q, t, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t+1))) \cdot \mathrm{rightphase}(Q, t, 1)$.

Proof. The goal follows from whitecenterphasematching(Q, t). □

Theorem 2.97. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k-1), 0) = g \wedge$headimportpath$(m \cdot P, start, b + (k-1)) \wedge$leftphase$(m \cdot P, b, k) +$ pathflux$(m \cdot P, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ centersign$(start, b + (k - 1) - start) \cdot$ rightphase$(m \cdot P, start, 2)$.

Proof. The goal follows from periodiccoupledcancellation(T, P, m, N, g). □

Theorem 2.98. *For every natural number Q and natural number b and natural number k, if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then* leftphase$(Q, b, k) +$ pathflux$(Q, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ leftgatephase$(Q, b + (k-1), 0)$.

Proof. The goal follows from leadingarrivalphaseeqgatephase(Q, b, k). □

Theorem 2.99. *For every natural number Q and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then* leftgatephase$(Q, finish, 0) =$ centersign$(start, finish - start) \cdot$rightphase$(Q, start, 2)$.

Proof. The goal follows from headimportpathgatephase$(Q, start, finish)$. □

Theorem 2.100. *For every natural number Q and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then not* leftgatephase$(Q, finish, 0) = -($centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2))$.

Proof. The goal follows from headimportpathrejectsnegativegate$(Q, start, finish)$. □

Theorem 2.101. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number start such that there exists a natural number*

b *such that there exists a natural number* k *such that* $N \leq$ *start and* $N \leq b \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{firstpatherasure}(m \cdot P, b, k, k-2) \wedge \mathrm{gatecode}(m \cdot P, b+(k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, \mathit{start}, b+(k-1)) \wedge \mathrm{leftgatephase}(m \cdot P, b+(k-1), 0) = \mathrm{centersign}(\mathit{start}, b+(k-1)-\mathit{start}) \cdot \mathrm{rightphase}(m \cdot P, \mathit{start}, 2)$.

Proof. The goal follows from $\mathrm{periodiccoupledgatephasesign}(T, P, m, N, g)$. □

Theorem 2.102. *For every natural number* T *and natural number* P *and natural number* m *and natural number* N*, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq N$*, then* $\mathrm{pathflux}(m \cdot P + m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$ *and* $\mathrm{pathflux}(m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = 0 \wedge \mathrm{pathflux}(m \cdot P, N + m \cdot P, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$.

Proof. The goal follows from $\mathrm{periodicdoubledseedfluxsplit}(T, P, m, N)$. □

Theorem 2.103. $\mathrm{leadingadvance}(11, 4, 4)$ *and* $\mathrm{centercleanthrough}(11, 4, 4) \wedge \mathrm{headimportpath}(11, 6, 7) \wedge \mathrm{centercleanthrough}(11, 6, 1) \wedge \mathrm{leftgatephase}(11, 7, 0) = 1 \wedge \mathrm{centersign}(6, 1) = 1 \wedge \mathrm{rightphase}(11, 6, 2) = 1$.

Proof. The goal follows from seedcoupledgatesign. □

Theorem 2.104. *For every natural number* Q *and natural number* t *and natural number* j*, if* $\mathrm{rightdefect}(Q, t+1, j+1) = \mathrm{true}$*, then* $\mathrm{rightdefect}(Q, t, j) = \mathrm{true}$ *or* $\mathrm{rightdefect}(Q, t, j+1) = \mathrm{true} \vee \mathrm{rightdefect}(Q, t, j+2) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{rightdefecthasancestor}(Q, t, j)$. □

Theorem 2.105. *For every natural number* Q *and natural number* t *and natural number* j *and natural number* T*, if* $1 \leq j$*, then if* $T \leq t$*, then if* $\mathrm{rightdefect}(Q, t, j) = \mathrm{true}$*, then if* $\mathrm{centercleanthrough}(Q, T, t-T)$*, then* $\mathrm{rightlineage}(Q, t, j, T)$.

Proof. The goal follows from $\mathrm{rightlineageofcleancenter}(Q, t, j, T)$. □

Theorem 2.106. *For every natural number* Q *and natural number* t *and natural number* k *and natural number* T*, if* $1 \leq k$*, then if* $T \leq t$*, then if* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$*, then if* $\mathrm{centercleanthrough}(Q, T, t-T)$*, then* $\mathrm{leftlineage}(Q, t, k, T)$.

Proof. The goal follows from $\mathrm{leftlineageofcleancenter}(Q, t, k, T)$. □

Theorem 2.107. *For every natural number Q and natural number t and natural number j and natural number T, if $T \leq t$, then if* $\mathrm{rightlineage}(Q, t, j, T)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq j + (t - T) \wedge j \leq i + (t - T) \wedge \mathrm{rightdefect}(Q, T, i) = \mathrm{true}$.*

Proof. The goal follows from $\mathrm{rightlineageatstart}(Q, t, j, T)$. □

Theorem 2.108. *For every natural number Q and natural number t and natural number k and natural number T, if $T \leq t$, then if* $\mathrm{leftlineage}(Q, t, k, T)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq k + (t - T) \wedge k \leq i + (t - T) \wedge \mathrm{leftdefect}(Q, T, i) = \mathrm{true}$.*

Proof. The goal follows from $\mathrm{leftlineageatstart}(Q, t, k, T)$. □

Theorem 2.109. *For every natural number Q and natural number t and natural number j, if* $\mathrm{rightlineage}(Q, t, j, 0)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq Q \wedge i \leq j + t \wedge j \leq i + t \wedge \mathrm{rightdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{rightcell}(Q, i) = \mathrm{true} \wedge \mathrm{rightphase}(Q, 0, i) = 1$.*

Proof. The goal follows from $\mathrm{rightlineageseed}(Q, t, j)$. □

Theorem 2.110. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftlineage}(Q, t, k, 0)$*, then there exists a natural number i such that $1 \leq i$ and $i \leq Q \wedge i \leq k + t \wedge k \leq i + t \wedge \mathrm{leftdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{leftcell}(Q, i) = \mathrm{true} \wedge \mathrm{leftphase}(Q, 0, i) = 1$.*

Proof. The goal follows from $\mathrm{leftlineageseed}(Q, t, k)$. □

Theorem 2.111. *For every natural number Q and natural number t, if $0 < Q$, then there exists a natural number y such that $y \leq t + Q$ and* $\mathrm{rightfront}(Q, t, y)$*.*

Proof. The goal follows from $\mathrm{existsrightfront}(Q, t)$. □

Theorem 2.112. *For every natural number Q and natural number t and natural number y and natural number z, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then $y \leq z + 1$.*

Proof. The goal follows from $\mathrm{rightfrontlower}(Q, t, y, z)$. □

Theorem 2.113. *For every natural number Q and natural number t and natural number y, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if $2 \leq y$, then* $\mathrm{rightdefect}(Q, t + 1, y - 1) = \mathrm{not}(\mathrm{rightcell}(t, y - 1))$*.*

Proof. The goal follows from $\mathrm{rightfrontdescent}(Q, t, y)$. □

Theorem 2.114. *For every natural number Q and natural number t and natural number y and natural number z, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $2 \leq y$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then if* $z + 1 \leq y$*, then* $z = y - 1$ *and* $\mathrm{rightcell}(t, y - 1) = \mathrm{false}$*.*

Proof. The goal follows from $\mathrm{rightfrontlockstep}(Q, t, y, z)$. □

Theorem 2.115. *For every natural number Q and natural number b and natural number k and natural number s, if* $\mathrm{rightmostleft}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $s+1 < k$*, then if* $\mathrm{rightdefect}(Q, b+s, 1) = \mathrm{true}$*, then* $\mathrm{center}(b + s) = \mathrm{true}$*.*

Proof. The goal follows from $\mathrm{leadingheadblackbeforearrival}(Q, b, k, s)$. □

Theorem 2.116. *For every natural number Q and natural number b and natural number k and natural number $start$, if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k - 1)$*, then if* $\mathrm{headimportpath}(Q, start, b + (k - 1))$*, then* $start = b + (k - 2)$*.*

Proof. The goal follows from $\mathrm{coupledimportbornatarrival}(Q, b, k, start)$. □

Theorem 2.117. *For every natural number Q and natural number b and natural number k and natural number s, if* $0 < Q$*, then if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k-1)$*, then if* $1 \leq s$*, then if* $s+1 < k$*, then there exists a natural number y such that* $\mathrm{rightdefect}(Q, b + s, 1) = \mathrm{false}$ *and* $\mathrm{rightfront}(Q, b + s, y) \wedge 2 \leq y \wedge y \leq k - s$*.*

Proof. The goal follows from $\mathrm{coupledrightfrontmirrorat}(Q, b, k, s)$. □

Theorem 2.118. *For every natural number Q and natural number b and natural number k and natural number s, if* $0 < Q$*, then if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k-1)$*, then if* $1 \leq s$*, then if* $s + 2 < k$*, then if* $\mathrm{rightfront}(Q, b + s, k - s)$*, then* $\mathrm{rightcell}(b + s, k - s - 1) = \mathrm{false}$ *and* $\mathrm{rightfront}(Q, b + (s + 1), k - (s + 1))$*.*

Proof. The goal follows from $\mathrm{coupledwhitestaircaseat}(Q, b, k, s)$. □

Theorem 2.119. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $g = 2$ *or* $g = 5$*, then there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that there exists a natural*

number y such that $N \leq start$ and $N \leq b \wedge T \leq b \wedge T \leq start \wedge 2 \leq k \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{gatecode}(m \cdot P, b + (k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, start, b + (k-1)) \wedge (3 \leq k \to start = b + (k-2)) \wedge (3 \leq k \to \mathrm{rightfront}(m \cdot P, b+1, y) \wedge 2 \leq y \wedge y \leq k-1) \wedge \mathrm{rightlineage}(m \cdot P, start, 2, T) \wedge \mathrm{leftlineage}(m \cdot P, b, k, T)$.

Proof. The goal follows from $\mathrm{periodiccoupledimportancestry}(T, P, m, N, g)$. □

Theorem 2.120. $\mathrm{leadingadvance}(14, 5, 3)$ *and* $\mathrm{centercleanthrough}(14, 5, 3) \wedge \mathrm{whitecentergap}(5, 2) \wedge \mathrm{center}(5) = \mathrm{true} \wedge \mathrm{rightfront}(14, 6, 2) \wedge \mathrm{headimportpath}(14, 6, 7)$.

Proof. The goal follows from seedcoupledimportatarrival. □

Theorem 2.121. *For every natural number t and natural number k and natural number n,* $\mathrm{bitvalue}(\mathrm{raybit}(t, k, n)) = \mathrm{bitvalue}(\mathrm{raybit}(t, k, 0)) + \mathrm{rayflux}(t, k, n)$.

Proof. The goal follows from $\mathrm{raybitbalance}(t, k, n)$. □

Theorem 2.122. *For every natural number Q and natural number d and natural number t and natural number p, if $d + p \leq t$, then* $\mathrm{leftphase}(Q + Q, t + (t - d - p), p) = \mathrm{diagonalphase}(Q, d, t) + \mathrm{diagonalflux}(Q, d, t, p)$.

Proof. The goal follows from $\mathrm{diagonalraybalance}(Q, d, t, p)$. □

Theorem 2.123. *For every natural number Q and natural number R and natural number d and natural number t and natural number p,* $\mathrm{diagonalflux}(Q + R, d, t, p) = \mathrm{diagonalflux}(Q, d, t, p) + \mathrm{diagonalflux}(R, d, t + Q, p)$.

Proof. The goal follows from $\mathrm{diagonalfluxlagadd}(Q, R, d, t, p)$. □

Theorem 2.124. *For every natural number Q and natural number z and natural number t, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(Q, z)$*, then if* $\mathrm{settledby}(z+1, t)$*, then* $\mathrm{diagonalphase}(Q, z+1, t) = -1 \vee \mathrm{diagonalphase}(Q, z+1, t) = 1$ *and* $\mathrm{diagonalphase}(Q, z+1, t+Q) = -\mathrm{diagonalphase}(Q, z+1, t) \wedge \mathrm{diagonalphase}(Q + Q, z+1, t) = 0$.

Proof. The goal follows from $\mathrm{diagonalphasedoubles}(Q, z, t)$. □

Theorem 2.125. *For every natural number Q and natural number z and natural number t, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{settledby}(z+2, t)$*, then* $\mathrm{diagonalphase}(Q, z+3, t+1) = -\mathrm{diagonalphase}(Q, z+1, t)$.

Proof. The goal follows from $\mathrm{diagonalresetphase}(Q, z, t)$. □

Theorem 2.126. *For every natural number Q and natural number z and natural number t, if* zerodiagonal(z)*, then if* settledby$(z+2,t)$*, then if* $z+3 \leq t$*, then* diagonalflux$(Q, z+3, t+1, 1) =$ diagonalflux$(Q, z+1, t, 1) + 2 \cdot$ diagonalphase$(Q, z+1, t)$.

Proof. The goal follows from diagonalresetfluxjump(Q, z, t). □

Theorem 2.127. *For every natural number T and natural number P and natural number m and natural number d and natural number t, if* centerperiodic(T, P)*, then if* $d+1 \leq t$*, then if* $T \leq t+(t-d-1)$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t+(t-d-1)+1))) \cdot$rightphase$(m \cdot P + m \cdot P, t+(t-d-1), 1)$.

Proof. The goal follows from periodicdiagonalwhiteheadbalance(T, P, m, d, t). □

Theorem 2.128. *For every natural number T and natural number P and natural number m and natural number d and natural number t and natural number start, if* centerperiodic(T, P)*, then if* $d+1 \leq t$*, then if* $T \leq start$*, then if* headimportpath$(m \cdot P + m \cdot P, start, t+(t-d-1))$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t + (t-d-1) + 1))) \cdot$ (centersign$(start, t + (t-d-1) - start) \cdot$ rightphase$(m \cdot P + m \cdot P, start, 2))$.

Proof. The goal follows from periodicdiagonalimportbalance$(T, P, m, d, t, start)$. □

Theorem 2.129. *For every natural number Q and natural number b and natural number k and natural number start and natural number q, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq q$*, then if* $q + 1 < k$*, then* rightfront$(Q, b+1, k-1)$ *and* rightcell$(b+1, q) =$ false $\wedge$ rightcell$(b+1+Q, q) =$ false.

Proof. The goal follows from coupledrightfrontexact$(Q, b, k, start, q)$. □

Theorem 2.130. *For every natural number Q and natural number b and natural number k and natural number start and natural number i, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq i$*, then if* $i + 1 < k$*, then* rightfront$(Q, b+i, k-i)$ *and* rightcell$(b+i, k-i-1) =$ false $\wedge$ rightcell$(b+i+Q, k-i-1) =$ false.

Proof. The goal follows from coupledforcedwhitestaircase$(Q, b, k, start, i)$. □

Theorem 2.131. *For every natural number Q and natural number b and natural number start, if* leadingadvance$(Q, b, 2)$*, then if* centercleanthrough$(Q, b, 2)$*, then if* headimportpath$(Q, start, b+1)$*, then* center$(b) =$ true *and* center$(b+1) =$ false $\wedge$($start < b \wedge$rightdefect$(Q, b, 1) =$ true $\wedge$ rightimportdifference$(Q, b) =$ false $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$bitvalue(center$(b+2))) \cdot$rightphase$(Q, b, 1) \vee$ $start = b \wedge$ rightdefect$(Q, b, 1) =$ false $\wedge$ rightcell$(b, 1) =$ false $\wedge$ rightcell$(b + Q, 1) =$ false $\wedge$ rightdefect$(Q, b, 2) =$ true $\wedge$ rightimportdifference$(Q, b) =$ true $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$bitvalue(center$(b+2))) \cdot$rightphase$(Q, b, 2))$.

Proof. The goal follows from depthtwocancellationcases$(Q, b, start)$. □

Theorem 2.132. *For every natural number Q and natural number b and natural number k and natural number start, if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then* $start = b+(k-2)$ *or* $k = 2$ *and* $start < b \wedge$rightdefect$(Q, b, 1) =$ true.

Proof. The goal follows from coupledimportbirthordepthtwosurvival$(Q, b, k, start)$. □

Theorem 2.133. *For every natural number Q and natural number b and natural number k and natural number start and natural number q, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then if* center$(b) =$ true*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $q+2 < k$*, then* rightcell$(b, q) =$ true *and* rightcell$(b+Q, q) =$ true.

Proof. The goal follows from coupledblackrootprefix$(Q, b, k, start, q)$. □

Theorem 2.134. leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ headimportpath$(4, 2, 6) \wedge$rightdefect$(4, 5, 1) =$ true $\wedge$ rightimportdifference$(4, 5) =$ false.

Proof. The goal follows from seeddepthtwosurvival. □

Theorem 2.135. *For every natural number Q and natural number t,* alignedbackground$(Q, t+1) =$ wittphi(alignedbackground(Q, t)).

Proof. The goal follows from alignedbackgroundstep(Q, t). □

Theorem 2.136. *For every natural number Q and natural number t,* $\mathrm{aligneddefect}(Q, t+1) = \mathrm{defectstep}(\mathrm{alignedbackground}(Q, t), \mathrm{aligneddefect}(Q, t))$.

Proof. The goal follows from $\mathrm{aligneddefectstep}(Q, t)$. □

Theorem 2.137. *For every natural number Q and natural number t,* $\mathrm{rowbit}(\mathrm{alignedbackground}(Q, t), 2 \cdot t + Q) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{backgroundtopbit}(Q, t)$. □

Theorem 2.138. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), 2 \cdot (t + Q)) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{aligneddefecttopbit}(Q, t)$. □

Theorem 2.139. *For every natural number Q and natural number t, if $0 < Q$, then* $\mathrm{twopow}(2 \cdot (t + Q)) \leq \mathrm{aligneddefect}(Q, t)$ *and* $\mathrm{aligneddefect}(Q, t) < \mathrm{twopow}(2 \cdot (t + Q) + 1)$.

Proof. The goal follows from $\mathrm{aligneddefectmagnitude}(Q, t)$. □

Theorem 2.140. *For every natural number Q and natural number t and natural number j, if $2 \cdot (t+Q) < j$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{aligneddefectabovetop}(Q, t, j)$. □

Theorem 2.141. *For every natural number Q and natural number t and natural number j, if $j < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{wittrow}(t+Q), j)$.

Proof. The goal follows from $\mathrm{aligneddefectlowprefix}(Q, t, j)$. □

Theorem 2.142. *For every natural number b and natural number d,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittlinear}(d), \mathrm{wittshl}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)), 1))$.

Proof. The goal follows from $\mathrm{defectsteplinear}(b, d)$. □

Theorem 2.143. *For every natural number b and natural number d,* $\mathrm{rowbit}(\mathrm{defectstep}(b, d), 0) = \mathrm{rowbit}(d, 0)$.

Proof. The goal follows from $\mathrm{defectsteplowbit}(b, d)$. □

Theorem 2.144. *For every natural number b and natural number d,* $\mathrm{vala}(\mathrm{defectstep}(b, d)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)))$ *and* $\mathrm{valb}(\mathrm{defectstep}(b, d)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(d$

Proof. The goal follows from $\mathrm{defectstepvalcoordinates}(b, d)$. □

Theorem 2.145. *For every natural number j,* $\mathrm{vala}(\mathrm{wittshl}(1, j+3)) = \mathrm{vala}(\mathrm{wittshl}(1, j))$ *and* $\mathrm{valb}(\mathrm{wittshl}(1, j+3)) = \mathrm{valb}(\mathrm{wittshl}(1, j)) \wedge (\mathrm{vala}(\mathrm{wittshl}(1, j)) = \mathrm{true} \vee \mathrm{valb}(\mathrm{wittshl}(1, j)) = \mathrm{true})$.

Proof. The goal follows from $\mathrm{edgevalcoordinates}(j)$. □

Theorem 2.146. *For every natural number t and natural number k,* $\mathrm{aligneddefect}(\mathrm{twopow}(k), t) = \mathrm{wittxor}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), \mathrm{wittxor}(\mathrm{wittrow}(t), \mathrm{wittshl}(\mathrm{wittrow}(t), \mathrm{twopow}(k) + \mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{aligneddefectwindow}(t, k)$. □

Theorem 2.147. *For every natural number t and natural number k, if* $t < \mathrm{twopow}(k)$, *then* $\mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), 0) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), 0) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), 2 \cdot (t + \mathrm{twopow}(k))) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), 2 \cdot (t + \mathrm{twopow}(k))) = \mathrm{false} \wedge \mathrm{vala}(\mathrm{wittwin}(t, \mathrm{twopow}(k))) = \mathrm{vala}(\mathrm{wittrow}(t + \mathrm{twopow}(k))) \wedge \mathrm{valb}(\mathrm{wittwin}(t, \mathrm{twopow}(k))) = \mathrm{valb}(\mathrm{wittrow}(t + \mathrm{twopow}(k))) \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), t + \mathrm{twopow}(k)) = \mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{dyadicwindowreadings}(t, k)$. □

Theorem 2.148. *For every natural number T and natural number P and natural number m and natural number t, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then* $\mathrm{rowbit}(\mathrm{wittphi}(\mathrm{aligneddefect}(m \cdot P, t)), t + m \cdot P + 1) = \mathrm{rowbit}(\mathrm{crosscarry}(\mathrm{alignedbackground}(m \cdot P, t), \mathrm{aligneddefect}(m \cdot P, t)), t + m \cdot P)$.

Proof. The goal follows from $\mathrm{periodiccentercrosscarry}(T, P, m, t)$. □

Theorem 2.149. $\mathrm{aligneddefect}(1, 0) = 5$ *and* $\mathrm{vala}(\mathrm{aligneddefect}(1, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(1, 0)) = \mathrm{true} \wedge \mathrm{aligneddefect}(1, 2) = 93 \wedge \mathrm{vala}(\mathrm{aligneddefect}(1, 2)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(1, 2)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(1, 2), 3) = \mathrm{true} \wedge \mathrm{aligneddefect}(4, 0) = 385 \wedge \mathrm{vala}(\mathrm{aligneddefect}(4, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(4, 0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(4, 0), 4) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{seedvalcancellation}$. □

Theorem 2.150. *For every natural number d and natural number n and natural number j, if* $j < n$ *or* $n + 2 < j$, *then* $\mathrm{rowbit}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n)), n) = \mathrm{not}(\mathrm{rowbit}(d, n))$ *and* $\mathrm{rowbit}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n)), j) = \mathrm{rowbit}(d, j) \wedge \mathrm{vala}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n))) = \mathrm{vala}(d) \wedge \mathrm{valb}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n))) = \mathrm{valb}(d)$.

Proof. The goal follows from $\mathrm{centerflipmaskinvisible}(d, n, j)$. □

Theorem 2.151. *For every natural number t and natural number j,* $\mathrm{rowbit}(\mathrm{observercode}(t), j) = \mathrm{leftcell}(t, j)$.

Proof. The goal follows from $\mathrm{observercodereadout}(t, j)$. □

Theorem 2.152. *For every natural number Q and natural number t,* $\mathrm{observerdefect}(Q, t) = \mathrm{wittxor}(\mathrm{observercode}(t + Q), \mathrm{observercode}(t))$.

Proof. The goal follows from $\mathrm{observerdefectalignment}(Q, t)$. □

Theorem 2.153. *For every natural number Q and natural number t and natural number j,* $\mathrm{rowbit}(\mathrm{observerdefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), t + Q + j)$.

Proof. The goal follows from $\mathrm{observerdefectreadout}(Q, t, j)$. □

Theorem 2.154. *For every natural number t,* $\mathrm{vala}(\mathrm{observercode}(t + 1)) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{bxor}(\mathrm{vala}(\mathrm{wittcarry}(\mathrm{observercode}(t))), \mathrm{observerinput}(t)))$ *and* $\mathrm{valb}(\mathrm{observercode}(t+1)) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{valb}(\mathrm{wittcarry}(\mathrm{observercode}(t))))$.

Proof. The goal follows from $\mathrm{observervalstep}(t)$. □

Theorem 2.155. *For every natural number T and natural number P and natural number m and natural number t, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then* $\mathrm{vala}(\mathrm{observerdefect}(m \cdot P, t+1)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t)))), \mathrm{leftdefect}(m \cdot P, t, 1))$ *and* $\mathrm{valb}(\mathrm{observerdefect}(m \cdot P, t+1)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t+m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t))))$.

Proof. The goal follows from $\mathrm{periodicobservervalstep}(T, P, m, t)$. □

Theorem 2.156. *For every natural number T and natural number P and natural number m and natural number t, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then* $\mathrm{rowbit}(\mathrm{observerdefect}(m \cdot P, t), t + m \cdot P) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{observerdefect}(m \cdot P, t), 0) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{periodicobserveredges}(T, P, m, t)$. □

Theorem 2.157. $\mathrm{observerdefect}(2, 0) = 7$ *and* $\mathrm{vala}(\mathrm{observerdefect}(2, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{observerdefect}(2, 0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(2, 0), 0) = \mathrm{true} \wedge \mathrm{observerdefect}(4, 0) = 24 \wedge \mathrm{vala}(\mathrm{observerdefect}(4, 0)) = \mathrm{true} \wedge \mathrm{valb}(\mathrm{observerdefect}(4, 0)) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(4, 0), 0) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{observerseedzeroval}$. □

3 Cyclotomic observer transport

Theorem 3.1. *For every natural number r and natural number a,* $\mathrm{qdivides}(r, \mathrm{wittlinpow}(r, a))$.

Proof. The goal follows from $\mathrm{qdividesintro}(r, a)$. □

Theorem 3.2. *For every natural number r and natural number d, if* $\mathrm{qdivides}(r, d)$, *then there exists a natural number a such that* $\mathrm{wittlinpow}(r, a) = d$.

Proof. The goal follows from $\mathrm{qdivideselim}(r, d)$. □

Theorem 3.3. *For every natural number r and natural number d and natural number n,* $\mathrm{qdivides}(r, \mathrm{wittshl}(d, n))$ *if and only if* $\mathrm{qdivides}(r, d)$.

Proof. The goal follows from $\mathrm{qdividesshl}(r, d, n)$. □

Theorem 3.4. *For every natural number r and natural number d, if* $0 < r$, *then if* $\mathrm{qdivides}(r, d)$, *then* $\mathrm{vala}(d) = \mathrm{false}$ *and* $\mathrm{valb}(d) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{qdividesval}(r, d)$. □

Theorem 3.5. *For every natural number r and natural number d, if* $d \neq 0$, *then if* $\mathrm{qdivides}(r, d)$, *then* $2 \cdot r \leq \mathrm{binarydegree}(d)$.

Proof. The goal follows from $\mathrm{qdividesdegreebound}(r, d)$. □

Theorem 3.6. *For every natural number d, if* $d \neq 0$, *then there exists a natural number r such that* $\mathrm{qdivides}(r, d)$ *and not* $\mathrm{qdivides}(r + 1, d)$.

Proof. The goal follows from $\mathrm{qorderexists}(d)$. □

Theorem 3.7. *For every natural number r and natural number d, if* $d \neq 0$, *then* $\mathrm{qdivides}(r, d)$ *if and only if* $r \leq \mathrm{qorder}(d)$.

Proof. The goal follows from $\mathrm{qdividesifforder}(r, d)$. □

Theorem 3.8. *For every natural number Q and natural number t,* $\mathrm{observersourceflux}(Q, t, 0) = 0$.

Proof. The goal follows from $\mathrm{observersourcefluxzero}(Q, t)$. □

Theorem 3.9. *For every natural number Q and natural number t and natural number n,* $\mathrm{observersourceflux}(Q, t, n{+}1) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{ob}$ $n), n + 1))$.

Proof. The goal follows from $\mathrm{observersourcefluxstep}(Q, t, n)$. □

Theorem 3.10. *For every natural number Q and natural number t and natural number n and natural number s,* observersourceflux$(Q, t, n + s) =$ wittxor(wittlinpow$(s,$ observersourceflux$(Q, t, n)),$ wittshl(observersourceflux$(Q, t+ n, s), n)$).

Proof. The goal follows from observersourcefluxappend(Q, t, n, s). □

Theorem 3.11. *For every natural number Q and natural number t,* observerqrank$(Q, t) = t + Q - 2 \cdot$ qorder(observerdefect(Q, t)).

Proof. The goal follows from observerqrankdefinition(Q, t). □

Theorem 3.12. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+1), 1) =$ wittxor(wittlinear(observerdefect$(m \cdot P, t)),$ wittshl(observersource$(m \cdot P, t), 1)$).

Proof. The goal follows from periodicobserverpolynomialstep(T, P, m, t). □

Theorem 3.13. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode$(t+ m \cdot P)),$ wittcarry(observercode$(t))),$ naturalbit(leftdefect$(m \cdot P, t, 1))$).

Proof. The goal follows from periodicobserversource(T, P, m, t). □

Theorem 3.14. *For every natural number T and natural number P and natural number m and natural number t and natural number n, if* centerperiodic(T, P)*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+n), n) =$ wittxor(wittlinpow$(n,$ observerdefect$(m \cdot P, t)),$ observersourceflux$(m \cdot P, t, n)$).

Proof. The goal follows from periodicobserverpolynomialwindow(T, P, m, t, n). □

Theorem 3.15. *For every natural number T and natural number P and natural number m and natural number t and natural number n and natural number r and natural number k, if* centerperiodic(T, P)*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then if $k \leq n+r$, then* qdivides$(k,$ observerdefect$(m \cdot P, t + n))$ *if and only if* qdivides$(k,$ observersourceflux$(m \cdot P, t, n))$.

Proof. The goal follows from periodicobserverwindowdivisibility(T, P, m, t, n, r, k). □

Theorem 3.16. *For every natural number T and natural number P and natural number m and natural number t and natural number r, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then if* $\mathrm{qdivides}(r, \mathrm{observerdefect}(m \cdot P, t))$, *then* $\mathrm{qdivides}(r + 1, \mathrm{observerdefect}(m \cdot P, t+1))$ *if and only if* $\mathrm{qdivides}(r+1, \mathrm{observersource}(m \cdot P, t))$.

Proof. The goal follows from $\mathrm{periodicobserverqgain}(T, P, m, t, r)$. □

Theorem 3.17. *For every natural number Q and natural number t, if* $0 < Q$, *then* $2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t)) \leq t + Q$.

Proof. The goal follows from $\mathrm{observerqorderbound}(Q, t)$. □

Theorem 3.18. *For every natural number T and natural number P and natural number m and natural number t and natural number n, if* $0 < P$, *then if* $0 < m$, *then if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then if* $0 < n$, *then* $\mathrm{observerqrank}(m \cdot P, t + n) < \mathrm{observerqrank}(m \cdot P, t)$ *if and only if* $\mathrm{qdivides}(\mathrm{qorder}(\mathrm{observerdefect}(m \cdot P, t)) + n/2 + 1, \mathrm{observersourceflux}(m \cdot P, t, n))$.

Proof. The goal follows from $\mathrm{periodicobserverqrankdrop}(T, P, m, t, n)$. □

Theorem 3.19. $\mathrm{observerdefect}(5, 2) = 216$ *and* $\mathrm{observerdefect}(5, 3) = 412 \wedge \mathrm{qdivides}(1, \mathrm{observerdefect}(5, 2)) \wedge \neg\, \mathrm{qdivides}(1, \mathrm{observerdefect}(5, 3)) \wedge \mathrm{center}(2) = \mathrm{false} \wedge \mathrm{center}(7) = \mathrm{false} \wedge \mathrm{center}(3) = \mathrm{true} \wedge \mathrm{center}(8) = \mathrm{true}$.

Proof. The goal follows from observerseedqloss. □

Theorem 3.20. *For every natural number r and natural number a and natural number d, if* $\mathrm{wittlinpow}(r, a) = d$, *then* $a \leq d$.

Proof. The goal follows from $\mathrm{qquotientbound}(r, a, d)$. □

Theorem 3.21. *For every natural number d and natural number r, if* $\mathrm{qdivides}(r+1, d)$, *then* $\mathrm{qordersearch}(d, r + 1) = r + 1$.

Proof. The goal follows from $\mathrm{qordersearchhit}(d, r)$. □

Theorem 3.22. *For every natural number d and natural number r, if not* $\mathrm{qdivides}(r + 1, d)$, *then* $\mathrm{qordersearch}(d, r + 1) = \mathrm{qordersearch}(d, r)$.

Proof. The goal follows from $\mathrm{qordersearchmiss}(d, r)$. □

Theorem 3.23. *For every natural number d,* $\mathrm{qorder}(d) = \mathrm{qordersearch}(d, \mathrm{binarydegree}(d))$.

Proof. The goal follows from qorderdefinition(d). □

Theorem 3.24. *For every natural number r and natural number d, if* qdivides(r, d), *then there exists a natural number a such that $a \leq d$ and* wittlinpow(r, a) = d.

Proof. The goal follows from qdividesboundedwitness(r, d). □

Theorem 3.25. *For every natural number d,* qordersearch($d, 0$) = 0.

Proof. The goal follows from qordersearchzero(d). □

Theorem 3.26. *For every natural number a and natural number y,* wittlinpow(twopow(a), y) = wittxor(wittxor(y, wittshl(y, twopow(a))), wittshl(y, twopow(a)+twopow(a))).

Proof. The goal follows from qfrobeniustwopow(a, y). □

Theorem 3.27. wittlinpow(16, 1) = wittxor(wittxor(1, wittshl(1, 16)), wittshl(1, 32)) *and* wittlinpow(16, wittxor(1, wittshl(1, 16))) = wittxor(1, wittshl(1, 48)).

Proof. The goal follows from qsixteenpage. □

Theorem 3.28. *For every natural number a and natural number y,* qdivides(twopow(a), wittxor(y, witts twopow(a)))).

Proof. The goal follows from qdividespageshifttwopow(a, y). □

Theorem 3.29. *For every natural number a and natural number n and natural number H,* qdivides(twopow(a), wittxor(H, wittxor(pagefold(3·twopow(a), H, n), pagetail(3· twopow(a), H, n)))).

Proof. The goal follows from qdividespagefoldtwopow(a, n, H). □

Theorem 3.30. *For every natural number L and natural number H and natural number p, if $0 < L$, then* page(L, wittcarry(H), p) = wittxor(wittcarry(page(L, H, p)), naturalbit(pa

Proof. The goal follows from pagecarry(L, H, p). □

Theorem 3.31. *For every natural number L and natural number H and natural number n, if $0 < L$, then* pagefold(L, wittcarry(H), n) = carryledger(L, H, n).

Proof. The goal follows from carryfoldeqledger(L, H, n). □

Theorem 3.32. *For every natural number a and natural number H and natural number n,* qdivides(twopow(a), wittxor(wittcarry(H), wittxor(carryledger(3· twopow(a), H, n), pagetail(3 · twopow(a), wittcarry(H), n)))).

Proof. The goal follows from carryledgercongruencetwopow(a, H, n). □

Theorem 3.33. *For every natural number T and natural number P and natural number m and natural number t and natural number a, if* centerperiodic(T, P), *then if $T \leq t$, then* qdivides(twopow(a), wittxor(observersource$(m \cdot P, t)$, wittxor(wittxor(carryledger(3·twopow(a), observercode$(t+m \cdot P)$, $t+m \cdot P+1)$, carryledger(3·twopow(a), observercode(t), $t+m \cdot P+1)$), naturalbit(leftdefect$(m \cdot P, t, 1)$))))).

Proof. The goal follows from periodicobserversourceledger(T, P, m, t, a). □

Theorem 3.34. *For every natural number T and natural number P and natural number m and natural number t and natural number r and natural number a, if* centerperiodic(T, P), *then if $T \leq t$, then if $r+1 \leq$* twopow(a), *then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$, *then* qdivides$(r+1,$ observerdefect$(m \cdot P, t+1))$ *if and only if* qdivides$(r+1,$ wittxor(wittxor(carryledger(3·twopow(a), observercode$(t+m \cdot P)$, $t+m \cdot P+1)$, carryledger$(3 \cdot$ twopow(a), observercode(t), $t+m \cdot P+1)$), naturalbit(leftdefect$(m \cdot P, t, 1)$))).

Proof. The goal follows from periodicobserverqgainledger(T, P, m, t, r, a). □

Theorem 3.35. qdivides(16, wittlinpow(16, 1)) *and* wittxor(wittphi(wittxor(2, wittlinpow(16, 1))), witt wittxor(wittlinpow(17, 1), 4)$\wedge\neg$ qdivides(16, wittxor(wittphi(wittxor(2, wittlinpow(16, 1))), wittphi(2)))

Proof. The goal follows from pageresiduenotclosed. □

Theorem 3.36. *For every natural number a and natural number $A0$ and natural number $A1$ and natural number $A2$,* qdivides(twopow(a), wittxor(wittxor(wittxor($A0$, wittshl($A1$ twopow(a))), wittxor(wittxor($A0, A2$), wittshl(wittxor($A1, A2$), twopow(a)))))).

Proof. The goal follows from qblockresidue$(a, A0, A1, A2)$. □

Theorem 3.37. *For every natural number a and natural number $A0$ and natural number $A1$ and natural number $A2$, if $A0 <$* twopow(twopow(a)), *then if $A1 <$* twopow(twopow(a)), *then if $A2 <$* twopow(twopow(a)), *then* qdivides(twopow(a), wittxor(wittxor($A0$, wittshl($A1$, twopow(a)))), wittshl($A2$, twopow$(a)+$ twopow(a)))) *if and only if $A0 = A2$ and $A1 = A2$.*

Proof. The goal follows from qblockchargezeroiff$(a, A0, A1, A2)$. □

Theorem 3.38. *For every natural number a and natural number A, if $A <$* twopow(twopow(a)), *then* wittcarry(wittxor(wittxor(A, wittshl(A, twopow(a)))), wittshl(A, twopow$(a)+$ twopow(a)))) = wittxor(wittxor(wittxor(wittcarry(A), wittshl(wittcarry(A), twopow(a)))), wittshl(witt twopow(a))), wittxor(wittshl(naturalbit(blockendpoint(twopow$(a), A)$), twopow(a))), wittshl(naturalbit twopow(a)))).

Proof. The goal follows from repeatedblockcarry(a, A). □

Theorem 3.39. *For every natural number a and natural number A, if* $A <$ twopow(twopow(a))*, then* qdivides(twopow(a), wittxor(wittcarry(wittxor(wittxor(A, wittshl(A, twopow(twopow(a))))), naturalbit(blockendpoint(twopow$(a), A$)))).

Proof. The goal follows from repeatedblockcarrycongruence(a, A). □

Theorem 3.40. *For every natural number a and natural number H and natural number p and natural number A, if* $A <$ twopow(twopow(a))*, then if* page(3·twopow$(a), H, p)$ = wittxor(wittxor(A, wittshl(A, twopow(a))), wittshl(A, twopow(a)+ twopow(a)))*, then* qdivides(twopow(a), wittxor(page(3·twopow(a), wittcarry$(H), p)$, wittxor(naturalbit twopow$(a), H, p)$))))).

Proof. The goal follows from pagecarryrepeated(a, H, p, A). □

Theorem 3.41. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if* $T \leq t$*, then* observersource$(m \cdot P, t)$ = wittxor(wittxor(wittcarry(observerdefect($m \cdot P, t$)), crosscarry(observercode(t), observerdefect($m \cdot P, t$))), naturalbit(leftdefect($m \cdot P, t, 1$))).

Proof. The goal follows from periodicobserversourcecrosscarry(T, P, m, t). □

Theorem 3.42. *There exists a natural number s such that there exists a natural number $s2$ such that* qdivides(16, wittxor$(s, s2)$) *and not* qdivides(16, wittxor(wittphi(s), wittphi$(s2$)

Proof. The goal follows from residuenotcongruence. □

Theorem 3.43. *For every natural number Q and natural number t and natural number j,* rowbit(observerdefect$(Q, t), j)$ = leftdefect(Q, t, j).

Proof. The goal follows from observerdefectleftreadout(Q, t, j). □

Theorem 3.44. *For every natural number r and natural number e and natural number k, if* $e \neq 0$*, then if* qdivides$(r,$ wittshl$(e, k))$*, then* $k + 2 \cdot r \leq$ binarydegree(wittshl(e, k)).

Proof. The goal follows from qfrontdegreebudget(r, e, k). □

Theorem 3.45. *For every natural number n and natural number e and natural number o, if* $e \neq 0$*, then if* twopow$(n) \leq o$*, then if* wittshl$(e, o) <$ twopow$(3 \cdot$ twopow$(n))$*, then not* qdivides(twopow(n), wittshl(e, o)).

Proof. The goal follows from qpageoffsetcharge(n, e, o). □

Theorem 3.46. *For every natural number Q and natural number t and natural number k, if $0 < Q$, then if* rightmostleft(Q, t, k)*, then* $k + 2 \cdot$ qorder(observerdefect$(Q, t)) \leq t + Q$.

Proof. The goal follows from observerfrontqbudget(Q, t, k). □

Theorem 3.47. *For every natural number Q and natural number t and natural number k, if $0 < Q$, then if* rightmostleft(Q, t, k)*, then* $k \leq$ observerqrank(Q, t).

Proof. The goal follows from observerfrontqrank(Q, t, k). □

Theorem 3.48. *For every natural number a and natural number Q and natural number t and natural number p and natural number o, if* twopow$(a) \leq o$*, then if $o < 3 \cdot$* twopow(a)*, then if* rightmostleft$(Q, t, 3 \cdot$ twopow$(a) \cdot p + o)$*, then not* qdivides(twopow(a), page$(3 \cdot$ twopow(a), observerdefect$(Q, t), p))$.

Proof. The goal follows from observerfrontpagecharge(a, Q, t, p, o). □

Theorem 3.49. *For every natural number Q and natural number t, if* leftdefect$(Q, t, 0) =$ false*, then* leftdefect$(Q, t + 1, 0) =$ bxor(leftdefect$(Q, t, 1)$, centergate$(Q, t))$.

Proof. The goal follows from centerdefectcoupling(Q, t). □

Theorem 3.50. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then* leftdefect$(m \cdot P, t, 1) =$ centergate$(m \cdot P, t)$.

Proof. The goal follows from periodiccentercoupling(T, P, m, t). □

Theorem 3.51. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then if* center$(t) =$ true*, then* leftdefect$(m \cdot P, t, 1) =$ false.

Proof. The goal follows from periodicblacknoarrival(T, P, m, t). □

Theorem 3.52. *For every natural number T and natural number P and natural number m and natural number t, if* centerperiodic(T, P)*, then if $T \leq t$, then if* center$(t) =$ false*, then* leftdefect$(m \cdot P, t, 1) =$ rightdefect$(m \cdot P, t, 1)$.

Proof. The goal follows from periodicwhitemirror(T, P, m, t). □

Theorem 3.53. *For every natural number T and natural number P, if* center$(T + P) =$ center(T)*, then if for every natural number t, if $T \leq t$, then* leftdefect$(P, t, 1) =$ centergate(P, t)*, then* centerperiodic(T, P).

Proof. The goal follows from centerperiodicofcoupling(T, P). □